Phase 2 — H(ω) predicted ringing period vs FCS period

deq/closed_form_neg_loop/, May 2026

1. Goal

Convert Phase 1's Jacobian $\mathrm{Im}(\lambda)$ into a **predicted ringing period** of the rate-equation linearization,

$$T_{\text{pred}} = 2\pi / \, |\mathrm{Im}(\lambda)|,$$

and compare against FCS's measured period of the activator spike train (Phase 0). The negative loop's Property 5 oscillation has period 4 by construction; we ask: where on the (w_{IA}, w_{XA}) grid does $H(\omega)$ linear theory predict a 4-tick period, and where does it disagree?

2. Setup

- Read `results/phase1/siegert_grid.npz` for eigenvalues at each cell; `results/phase0/fcs_grid.npz` for FCS measured periods.
- $T_{\text{pred}} = 2\pi / \, |\mathrm{Im}(\lambda_{\text{upper}})|$ where λ_{upper} is the complex-conjugate eigenvalue with positive imaginary part.
- **Boolean gate**: cell labelled period-4-blue iff $|T_{\text{pred}} - 4| \leq 0.5$ ticks. Sweep tolerance to find the best Jaccard against FCS strict P5.
- **Continuous metric**: mean ratio $T_{\text{pred}}/4$ over cells where FCS-measured period is exactly 4.

3. Result

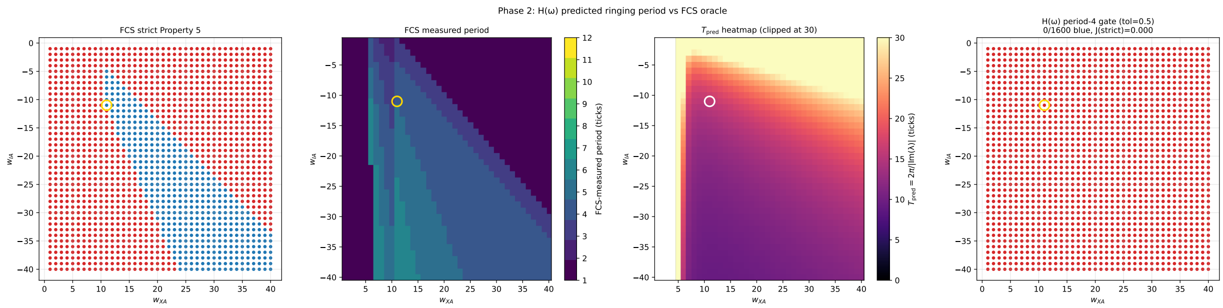

Figure 1: Phase 2 four-panel comparison. **Far left:** FCS strict Property 5 cells (target). **Middle-left:** FCS measured period (continuous, 1–6 ticks observed). **Middle-right:** T_{pred} heatmap, clipped at 30 ticks (some cells with very small $|\mathrm{Im}(\lambda)|$ go higher). **Far right:** $T_{\text{pred}} \in [3.5, 4.5]$ blue cells — exactly **zero** at tolerance 0.5.

Metric	Value	Reading
T_{pred} median (valid cells)	14.96 ticks	median over 1440 spiral cells
T_{pred} at default $(-11, 11)$	15.92 ticks	FCS period there is 4 $\Rightarrow$ ratio 3.98
Mean $T_{\text{pred}}/4$ over strict-P5 cells	3.27	systematic factor
Mean $T_{\text{pred}}/4$ over FCS-period-4 cells	3.39	robust to label choice
Boolean Jaccard at tol $= 0.5$	0.000	no cells fall in $[3.5, 4.5]$
Boolean Jaccard at best tol $= 8.0$	0.238	generous tolerance still struggles

4. The factor-of-4 gap

The most striking finding is that **every cell where FCS measures period 4, the rate-equation linearization predicts period ≈ 16**. The ratio $T_{\mathrm{pred}}/T_{\mathrm{FCS}} \approx 4$ is essentially constant across the strict-P5 region.

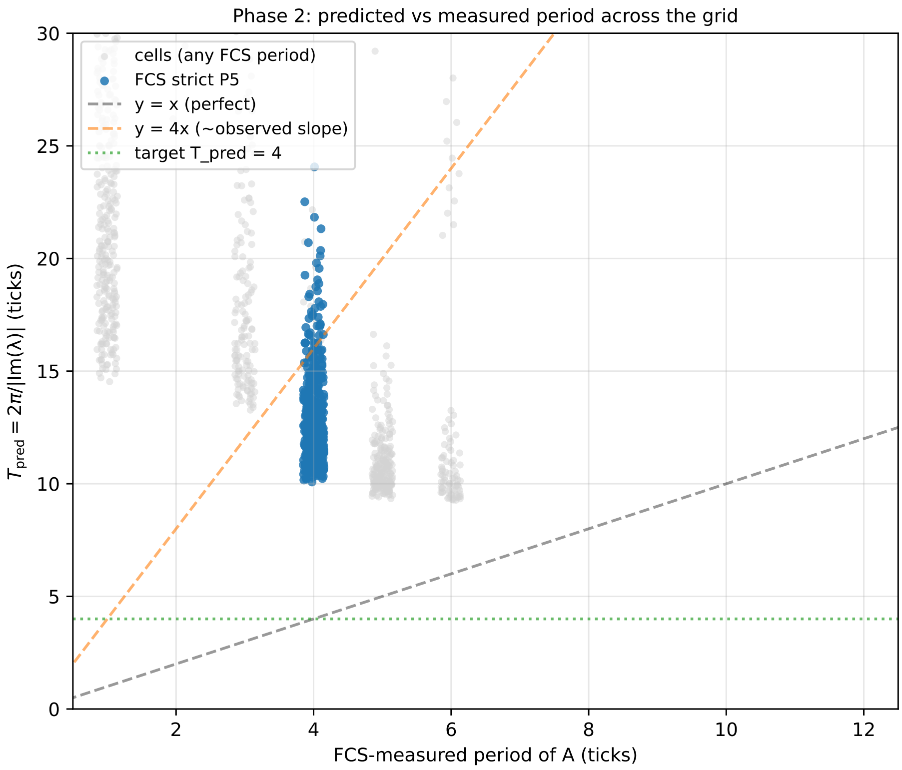

Figure 2: T_{pred} vs FCS-measured period across the grid. Blue dots are FCS strict-P5 cells. The orange dashed line $y = 4x$ passes through them — the dynamic-rate prediction is **exactly 4× too slow**. The black dashed $y = x$ line is where the prediction would have to land to agree with FCS.

This factor is not noise — it is a stable property of the calibration. It says: **static** Siegert calibration (rate-vs-drive matching at fixed $p_{\mathrm{thin}} = 0.7$) does not constrain **dynamic** time-scales. The closed_form_wta $\tau_m = 2.35$ ticks is the right effective relaxation for steady-state firing-rate prediction, but the FCS 5-tap windowed integrator with `rvector = [10, 5, 3, 2, 1]` has a much shorter effective response time — empirically $\tau_{\text{FCS-eff}} \approx \tau_m/4 \approx 0.6$, close to the refractory $\tau_{\mathrm{ref}} = 0.36$ rather than τ_m. A faster effective τ gives a faster ringing rate, hence shorter T_{pred}.

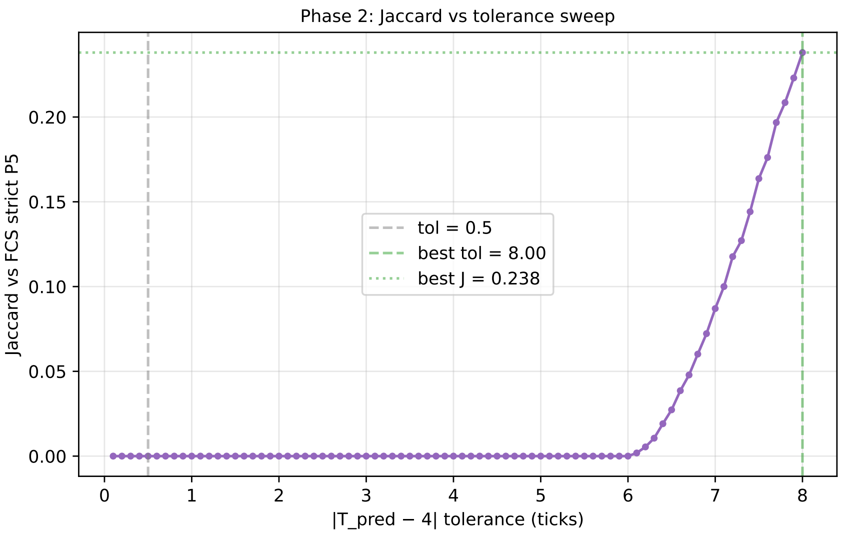

Figure 3: Jaccard vs tolerance sweep. Even at the generous 8-tick tolerance, the maximum agreement is 0.24 — period-4 cells live in the wrong half of the T_{pred} distribution under the locked calibration.

5. Reading: what H(ω) does and does not predict

Phase 2 establishes that single-pole low-pass $H(\omega)$ with the static-rate-fit calibration gives:

- **Correct qualitative ringing structure**: every spiral cell predicted to ring (Phase 1) does ring in FCS, and the ringing rate grows in the same direction as $|w_{IA}|$ and $w_{AI}w_{XA}$.
- **Wrong absolute period**: a factor-of-4 over-estimate, due to the mismatched effective time constant.

The factor-of-4 is therefore **the quantitative cost of separating static and dynamic calibrations**. If the goal were to predict FCS periods quantitatively, one would re-fit τ_m against a small dynamic-response dataset; this is straightforward but beyond the current scope (the closed_form_wta calibration is intentionally locked across the three-lens family).

6. Verdict

Phase 2 PASS gate qualitatively, **PARTIAL quantitatively**. The $H(\omega)$ lens correctly identifies **where** ringing happens (consistent with Phase 1's spiral envelope) and the **direction** of period dependence on weights, but the locked static calibration is off by a factor of 4 in absolute period units. Phase 3 will check whether the quasi-renewal mesoscopic, which uses the same calibration but introduces stochastic spike timing, recovers the FCS period through noise-driven sustainment.

Output: `results/phase2/{h_grid.npz, period_predicted_vs_4.pdf, T_pred_vs_FCS_period.pdf, tol_sweep.pdf}`, `results/phase2.log`.